

Improving Day-Ahead Energy Forecasts for Power System Operations with Open-Source Data and Machine Learning

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Net Demand Forecasting Errors are Growing

- ▶ Weather-dependent demand and generation is increasing net demand **variability**.
- ▶ This increase in net demand variability is causing **forecasting errors** to grow.

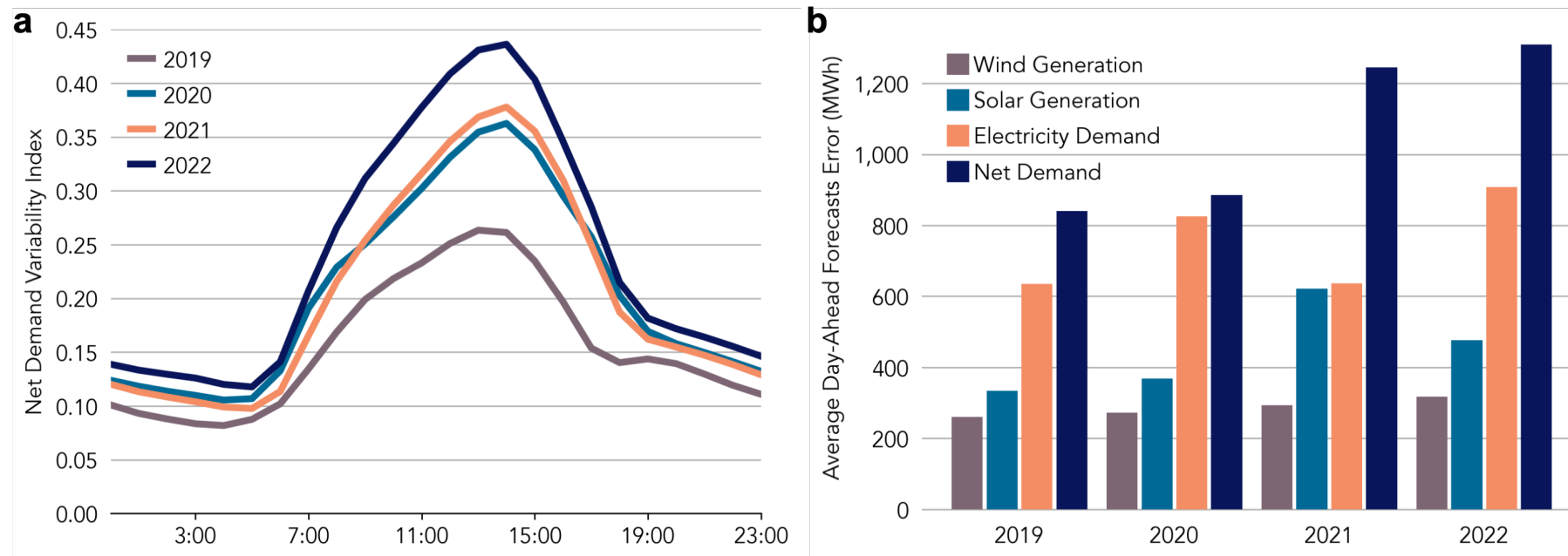


Figure 1: CAISO's net electricity demand variability index (a), and average forecasting errors (b) from 2019 to 2022; Source: www.caiso.com

Increasing Reserve Requirements and Imbalances

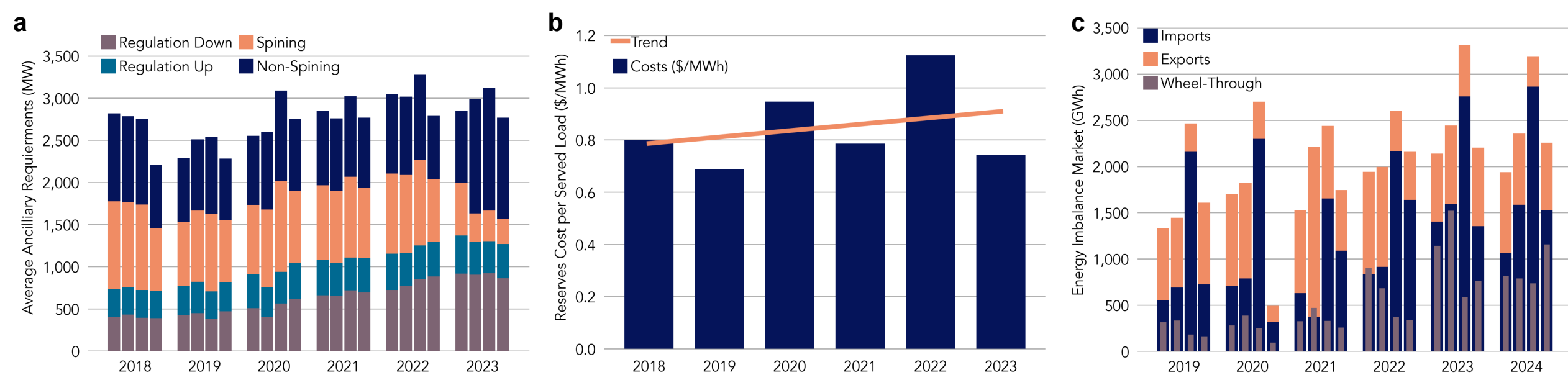
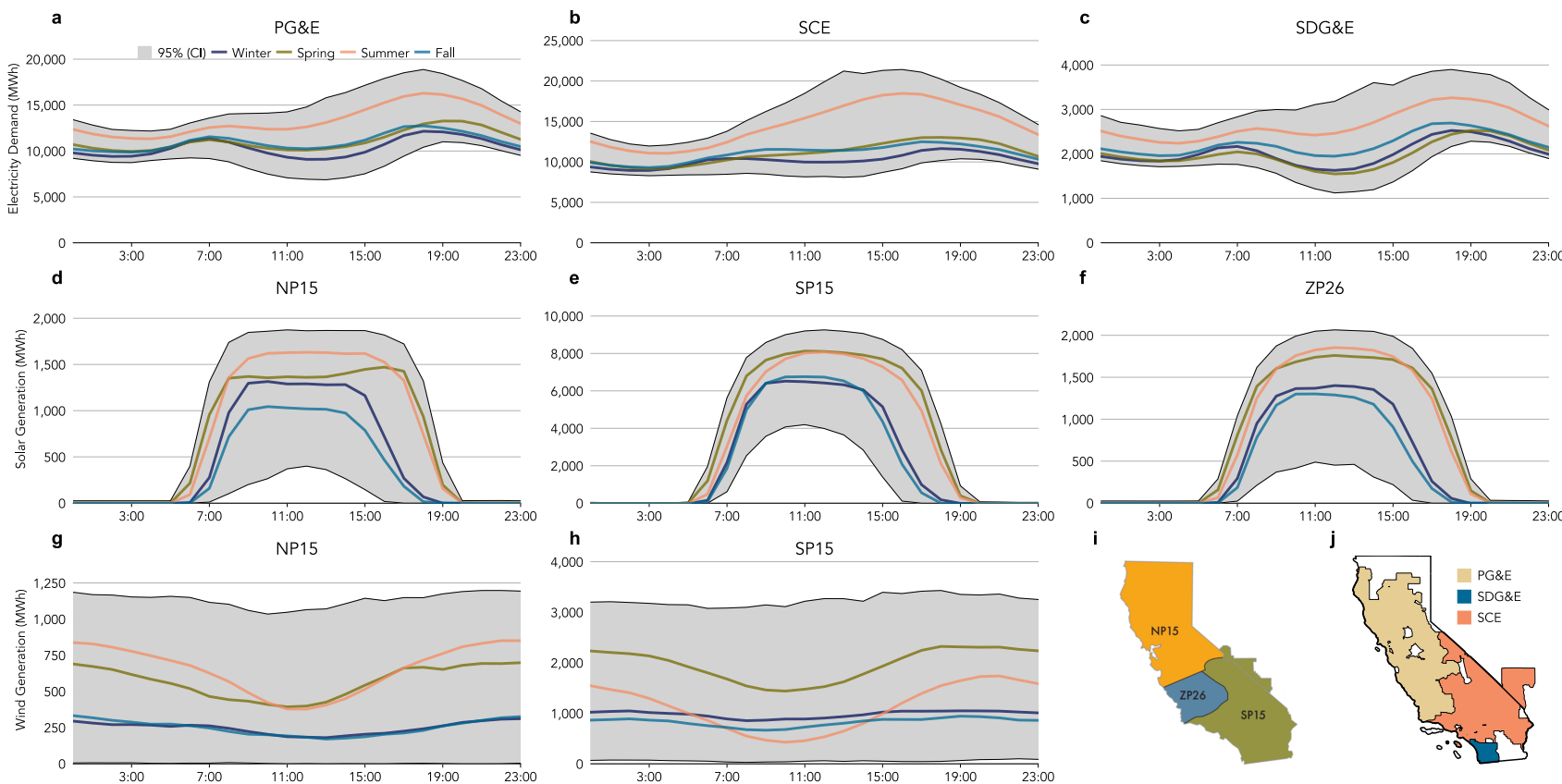


Figure 2: CAISO's operating reserves requirements (a), reserves costs and trend (b), and imports and exports (c); Source: www.caiso.com

- Growth in **forecasting errors** has led to an increase in:
 - **imports** and **exports** from the imbalance market,
 - **operating reserves** procured in the ancillary services market and their associated **costs**.

Electricity Demand and Supply Uncertainty



- **Demand** from the largest customer-serving utilities in California (PG&E, SDGE, and SCE).
- **Solar** at the northern (NP15), southern (SP15), and central (ZP26) trading hubs.
- **Wind** at NP15 and SP15 trading hubs.

Figure 3: 2022 CAISO's demand (a, b, c), solar (d, e, f) and wind (g, h) generation (95% confidence interval and seasonal averages), trading hubs (i) and utility (j) areas; Source: www.oasis.caiso.com

Forecast Operational Characteristics

- ▶ The proposed model assimilates:
 - ▶ **11 weather features** in operational day (d),
 - ▶ from High-Resolution Rapid Refresh (**HRRR**) Numerical Weather Prediction (NWP) **forecast** from NOAA at 16:00 ($t = 16$ interval),
 - ▶ to forecast hourly **demand, solar, and wind generation** in operational day (d).
- ▶ We used NWP from Jun 2019 to May 2022 for **training** and from Jun 2022 to Mar 2023 for **testing**.

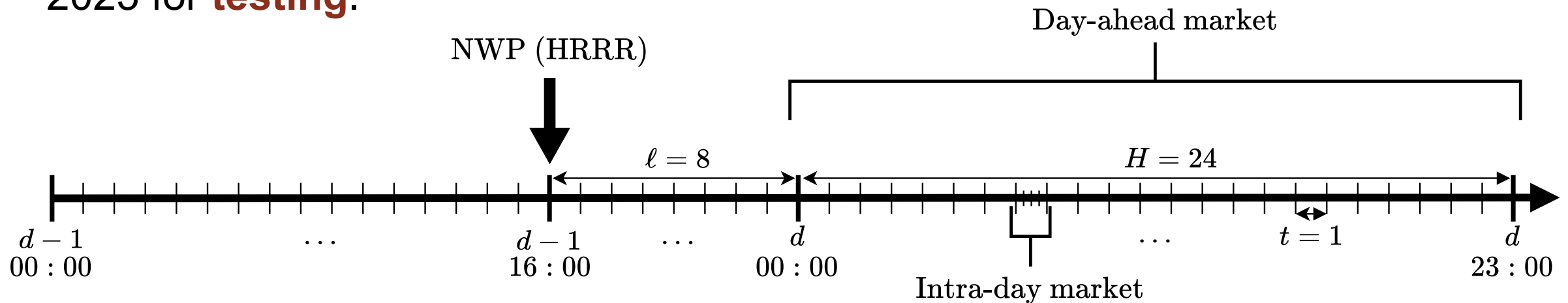
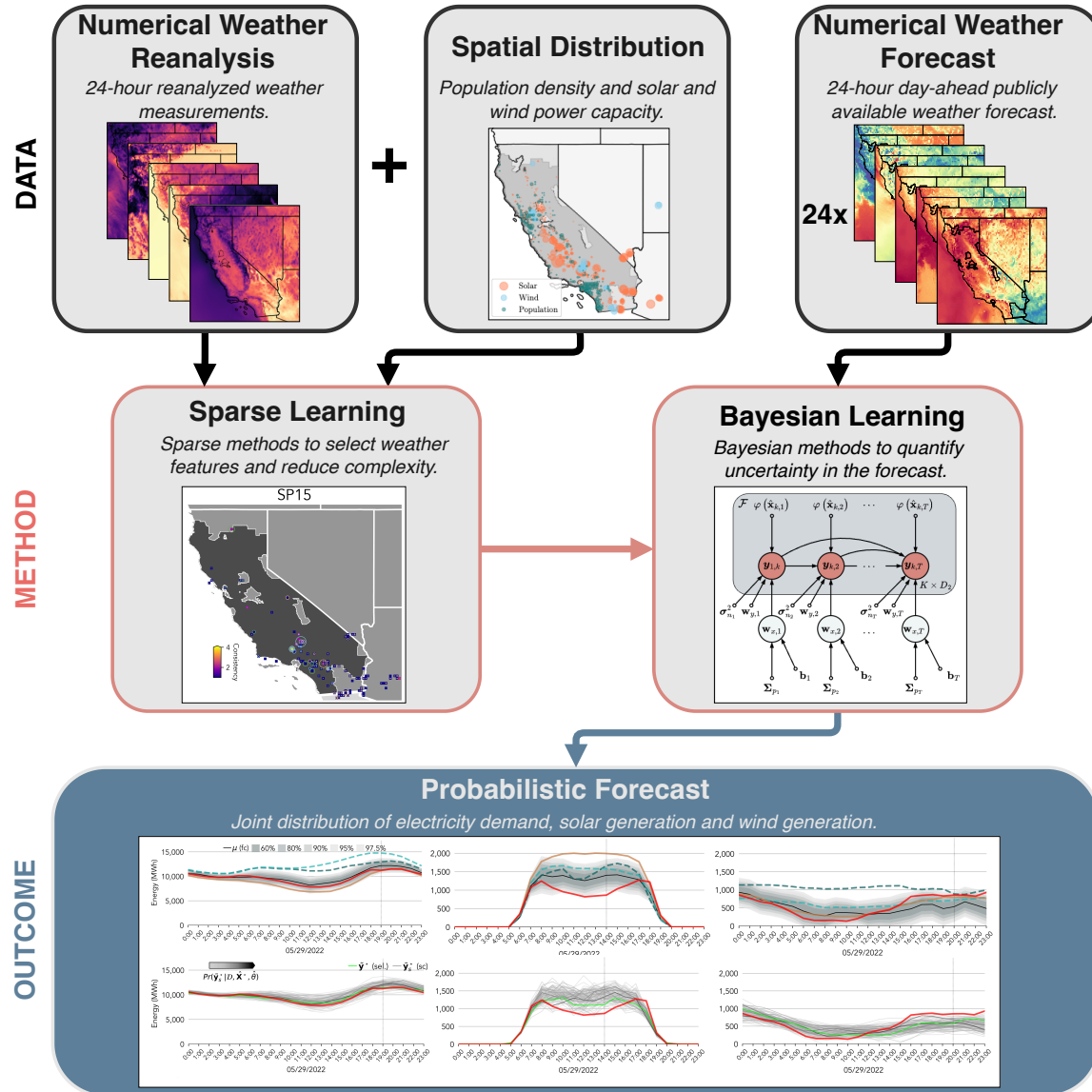


Figure 4: The proposed day-ahead forecast characteristics are $\ell = 8$ hours (lead time), $h = 24$ hours (horizon), and $t = 1$ hour (granularity).

Joint Day-Ahead Probabilistic Energy Forecast



1. **Open-source** Numerical Weather Forecast (NWF).
2. **Asset-level spatial filtering** to reduce dimensionality.
3. **Sparse learning** to select the weather features.
4. **Bayesian learning** to quantify uncertainty and generate scenarios.
5. **Chain of Models** to generate realistic scenarios.

Figure 5: Workflow.

Sparse Learning

Weather feature selection:

- **Lasso**: L_1 –norm regularization.
- **Orthogonal Matching Pursuit** (OMP): L_0 –norm.
- **Elastic Net** (EN): L_1 –norm and L_2 –norm.
- **Group Lasso** (GL): L_0 –norm and L_1 –norm.

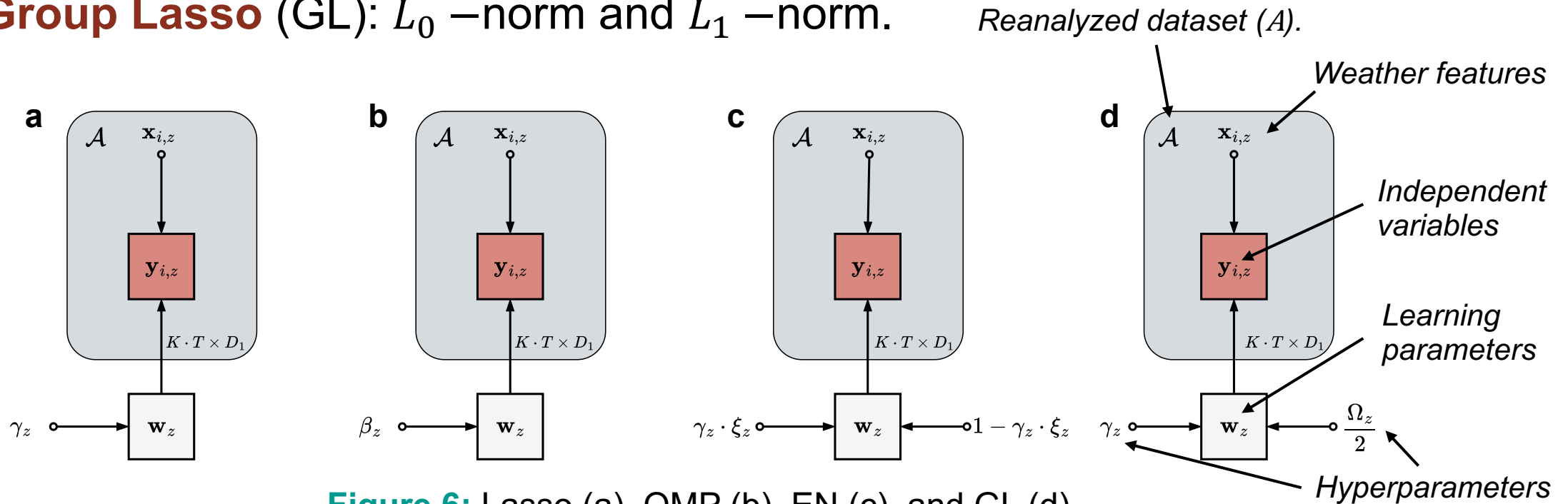


Figure 6: Lasso (a), OMP (b), EN (c), and GL (d).

Bayesian Learning

- **Bayesian Linear Regression**¹ (BLR): Independent models.
- **Relevance Vector Machine**² (RVM): BLR with automatic relevance determination.
- **Gaussian Process for Regression**¹ (GPR): Non-linear independent models (“kernel trick”).
- **Multi-Task GPR**³ (MTGPR): GPR joint at system-level (SLGPR), or at nodal-level (NLGPR).

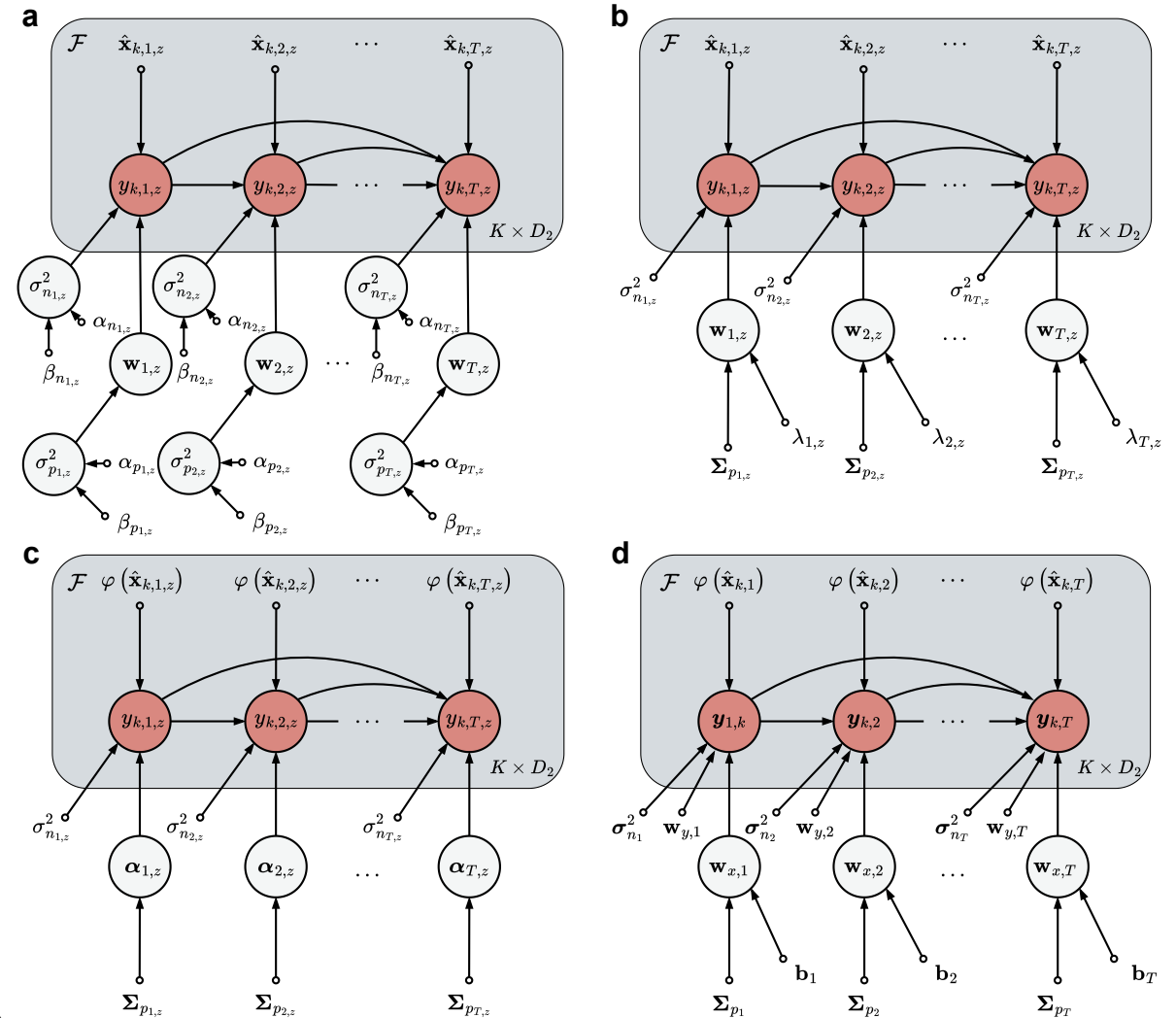


Figure 7: Bayesian inference: (a) BLR, (b) RVM, (c) GPR, and (d) MTGPR.

¹Rasmussen et al., Gaussian processes for machine learning (2006).

²Tipping, Sparse Bayesian learning and the relevance vector machine (2001).

³García-Hinde et al., A conditional one-output likelihood formulation for multitask Gaussian processes (2022).

Probabilistic Energy Forecasts are Competitive

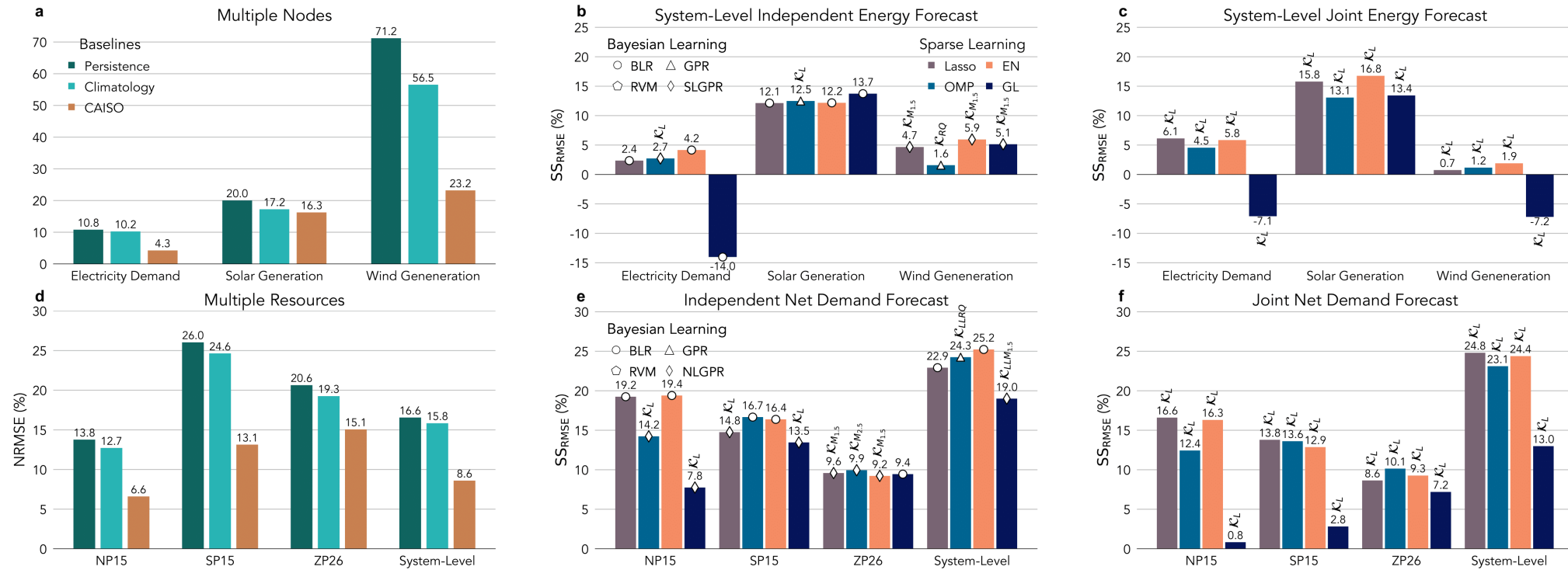
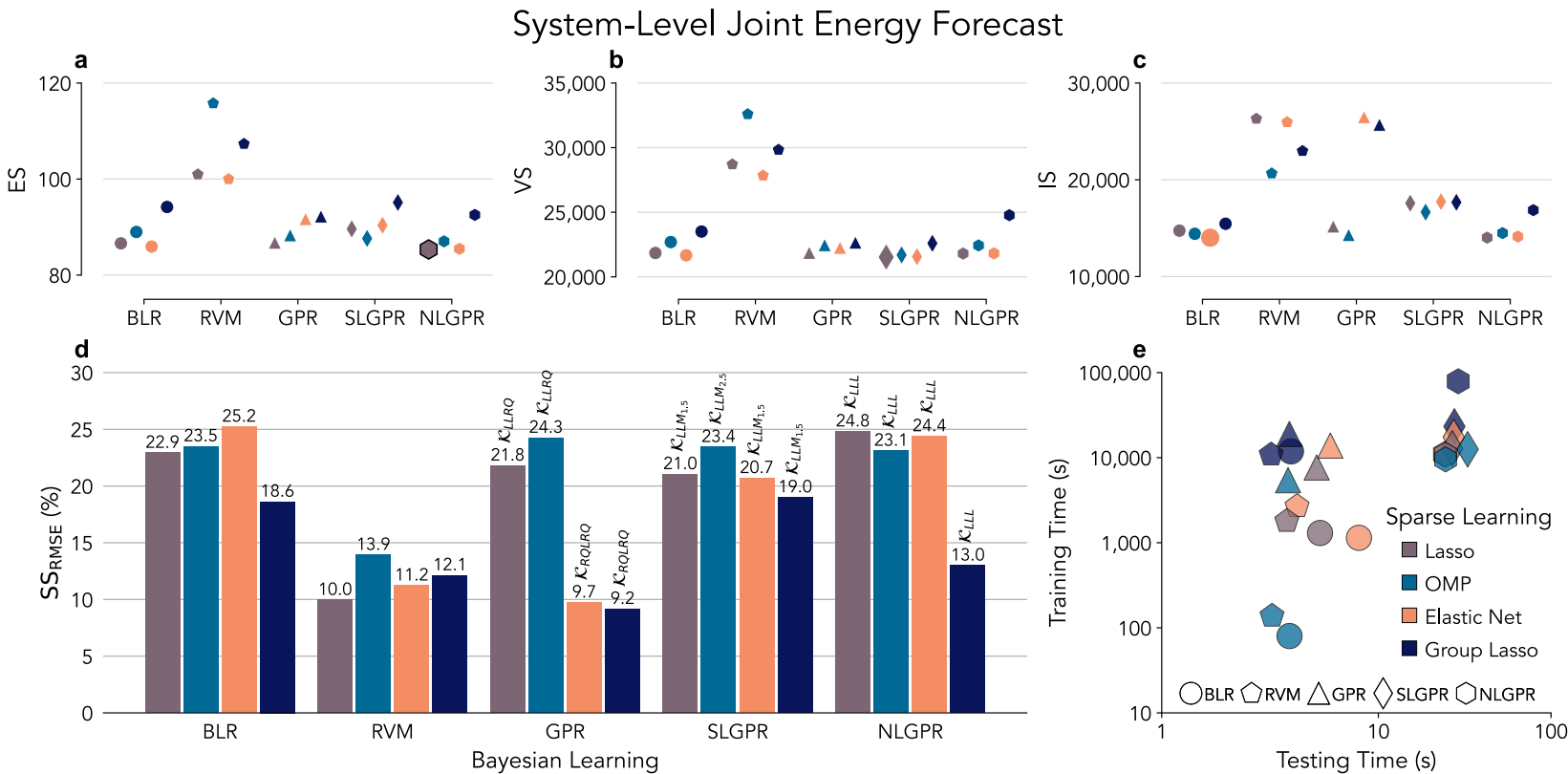


Figure 8: Baseline system-level forecast errors (a) and Skill Scores (SS) for independent (b) and joint (c) forecasts. Baseline nodal-level forecast errors (d) and SS for independent (e) and joint (f) net-demand forecasts.

- The proposed model **improve upon** point-wise forecasts from **baseline (CAISO)**.
- Joint forecasts **perform better than** independent forecasts at the **system-level**.
- Independent forecasts **perform better than** joint forecasts at the **nodal-level**.

Multiple Criteria for Model Selection



- Multivariate **proper scoring rules**⁴:
- **Energy Score (ES).**
- **Variogram Score (VS).**
- **Interval Score (IS)** - 60%, 80%, 90%, 95% and 97.5%.
- Each **proper scoring rules** evaluates a different **property!**

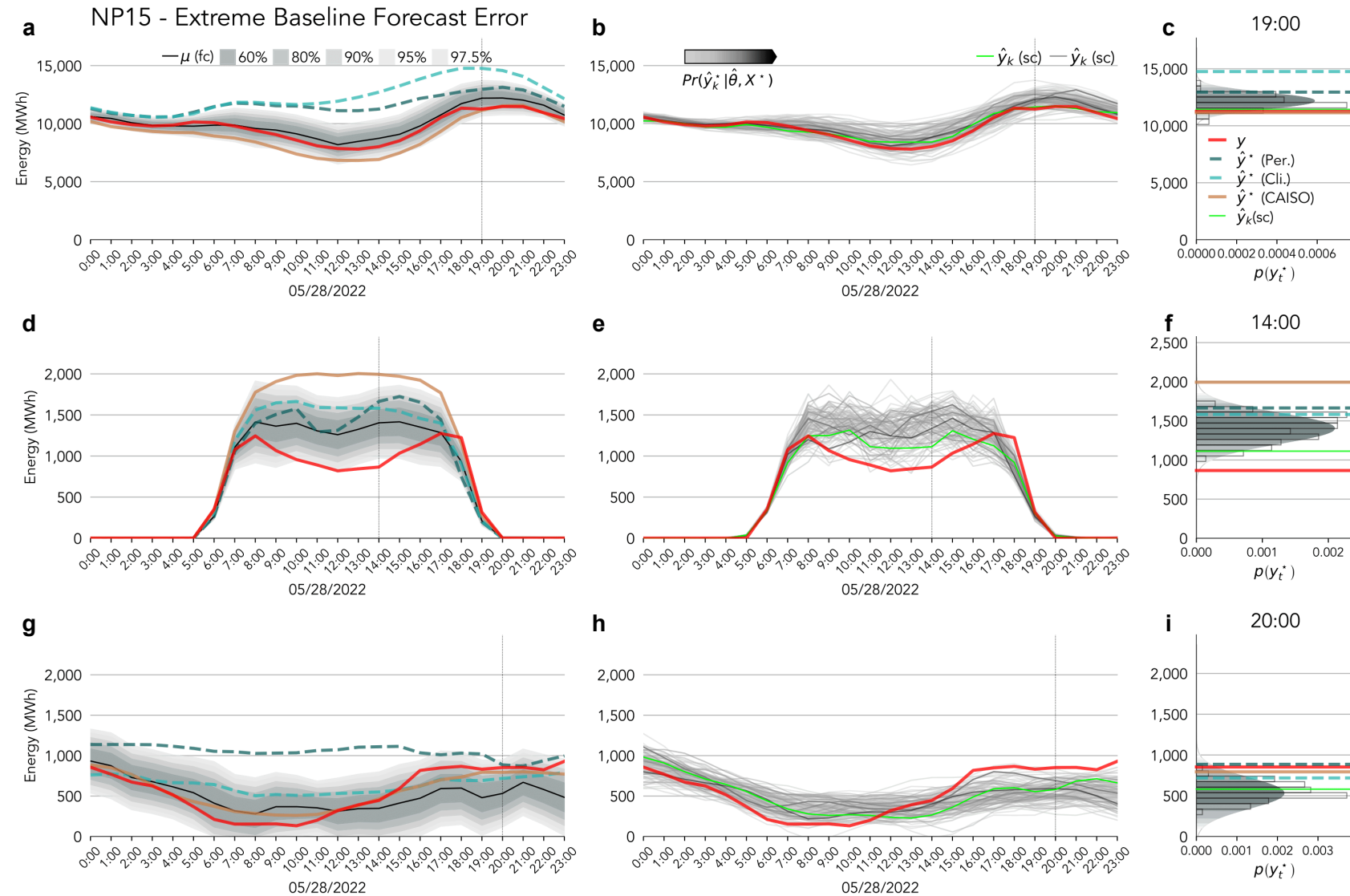
Figure 9: An accurate model (a and d, ↓ ES and ↑ SS_{RMSE}) that generates realistic scenarios (b, ↓ VS) from a calibrated predictive distribution (c, ↓ IS) with low computational cost (e, ↓ time) requires looking at multiple scores.

⁴Gneiting et al., Strictly proper scoring rules, prediction, and estimation (2007).

Robust Day-Ahead Forecast on Extreme Events

- **Joint** energy forecast at **nodal-level** (NP15).
- Extreme net demand forecasting error on **May 29, 2022**.

Figure 10: Joint day-ahead energy (demand, solar and wind generation) forecast density and scenarios. A joint scenario is highlighted.



Dynamic Reserves Allocation Reduces Imbalances

- Allocation (from **Jun 2022** to **Mar 2023**) based on **predictive distribution**:
 - by finding the **confidence level** containing a target reserve capacity,
 - reduces imports**, slightly increasing exports.
- Model selection based on **interval score reduces imbalances**.

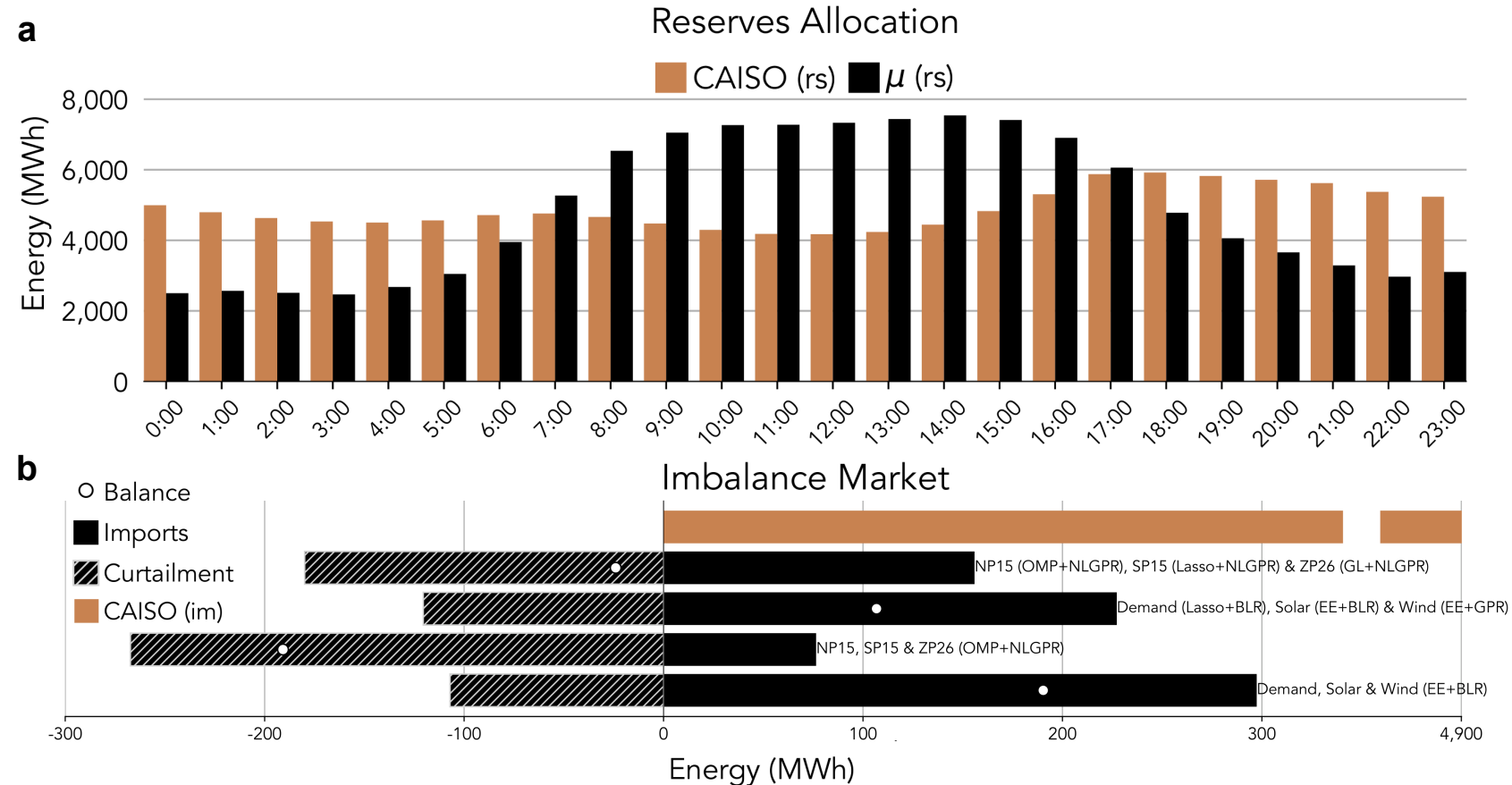


Figure 11: The proposed reserve allocation adapts to the uncertainty in the day-ahead net demand forecast.

Conclusions

Challenges:

- ▶ Quantify the **benefits** of a probabilistic forecast to incentivize adoption.
- ▶ Provide a **workflow** to incorporate a probabilistic forecast into electricity market operation.
- ▶ Assign **risk** to forecast scenarios based on their probability.

Take Aways:

A probabilistic day-ahead forecast based on **open-source** data:

- ▶ reach **similar accuracy** than ISOs' forecast baseline.
- ▶ has the potential to **dynamically** allocate reserves **more efficiently** in response to the uncertainty.

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